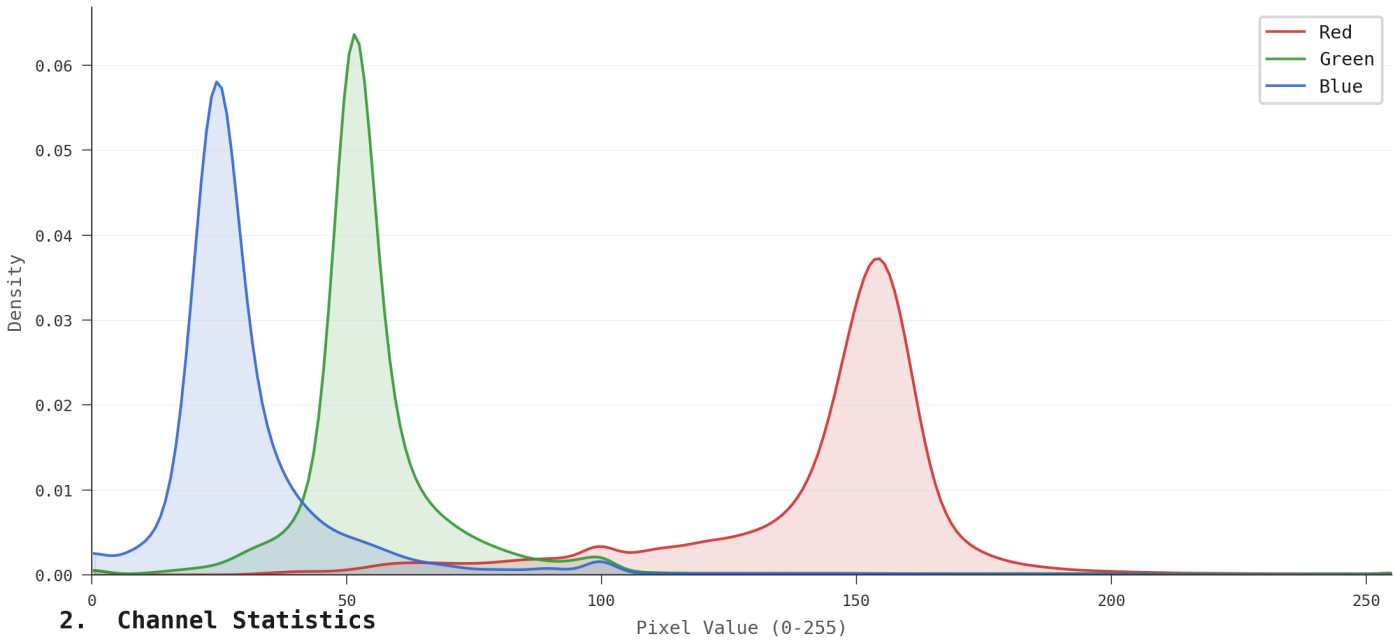
Reference Data Sheet: KODIM02

Subject: Red Door Latch and Bolt

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	142.8563	27.8074	773.2493	3.8056	-1.3758	0	255
Green	56.3640	21.6586	469.0935	30.1578	4.5648	0	255
Blue	31.2092	20.9966	440.8553	20.4990	3.8692	0	254

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.5279	0.4063
G	0.5279	1.0000	0.9740
B	0.4063	0.9740	1.0000

Pairwise Pearson Correlation Coefficients

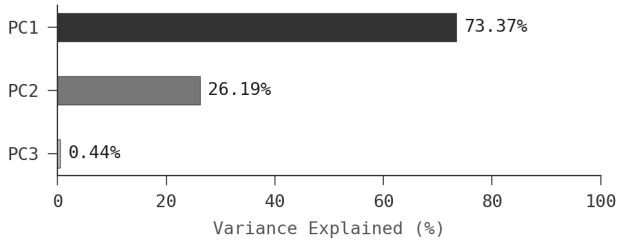
r(R-G)	=	**0.527910**
r(R-B)	=	**0.406272**
r(G-B)	=	**0.973974**
Avg |r|	=	**0.636052**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{773.2513} & 317.9436 & 237.2061 \\
317.9436 & \mathbf{469.0947} & 442.9210 \\
237.2061 & 442.9210 & \mathbf{440.8564}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM02

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	**-0.6501**	-0.5642	-0.5089	1234.9118
PC2	-0.7550	0.4043	0.5162	440.8250
PC3	0.0855	-0.7198	0.6889	7.4656

6. Derived Metrics

Condition Number

165.41

(l1/l3) PC1/PC2: 2.80 | PC2/PC3: 59.05

Blue Channel Independence

27.4%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Red

Loading: -0.6501 | Red dominant

Classification: Three-Dimensional (PC1 = 73.37%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.936186	0.916138	0.926162
Green	0.913504	0.884900	0.899202
Blue	0.907388	0.874535	0.890961

8. Average Local Variance (3x3)

Channel	Value
Red	91.0137
Green	68.9827
Blue	69.9345

9. Redundancy Profile

Classification: Three-Dimensional | Condition Number: 165.41 | PC1: 73.37%

Correlation: G–B pair dominates at r = 0.973974 | R–G: 0.527910 | R–B: 0.406272

Blue Channel: 27.4% independent | Substantial independence from PC1

Spatial: Avg autocorrelation: 0.905442 | Avg local variance: 76.6436

Eigenvector: Red dominant | R (-0.6501) leads | G (-0.5642), B (-0.5089)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."